UNDER-DASH POWER BOARD: block diagram. Each box is a sheet (double-click it in KiCad).
Power rails +12V, +5V, +3V3 and GND join the sheets by name (power symbols);
every other connection between sheets is a wire here.

12 V input and protection

File: input.kicad_sch

5 V converter (TPS54360, TI reference design)

File: buck.kicad_sch

Inter-box cable: DB-25 to the pod

5V_POD
BRAKE_RPWM
BRAKE_LPWM
BRAKE_EN
MAGNET_EN
BRAKE_CURRENT
RELAY_LEFT
RELAY_RIGHT
RELAY_HAZARD
RELAY_HORN
RELAY_BEAM
CAN_TX
CAN_RX
CAN_STBY
BOX_PRESENT
BRAKE_LIGHT
KILL_SENSE
HX711_DOUT
HX711_SCK

File: cable.kicad_sch

3.3 V, the pod's 5 V, the LIDAR feed

5V_POD

Brake actuator: buffer, BTS7960, current sense, cable magnet

BRAKE_RPWM
BRAKE_LPWM
BRAKE_EN
MAGNET_EN
BRAKE_CURRENT

File: actuator.kicad_sch

Signal relays: indicators, hazards, horn, high beam

RELAY_LEFT
RELAY_RIGHT
RELAY_HAZARD
RELAY_HORN
RELAY_BEAM

File: relays.kicad_sch

CAN transceiver for OBD-II

CAN_TX
CAN_RX
CAN_STBY

File: can.kicad_sch

Presence, brake light, E-stop sense, kill relay, bench

BOX_PRESENT
BRAKE_LIGHT
KILL_SENSE
HX711_DOUT
HX711_SCK

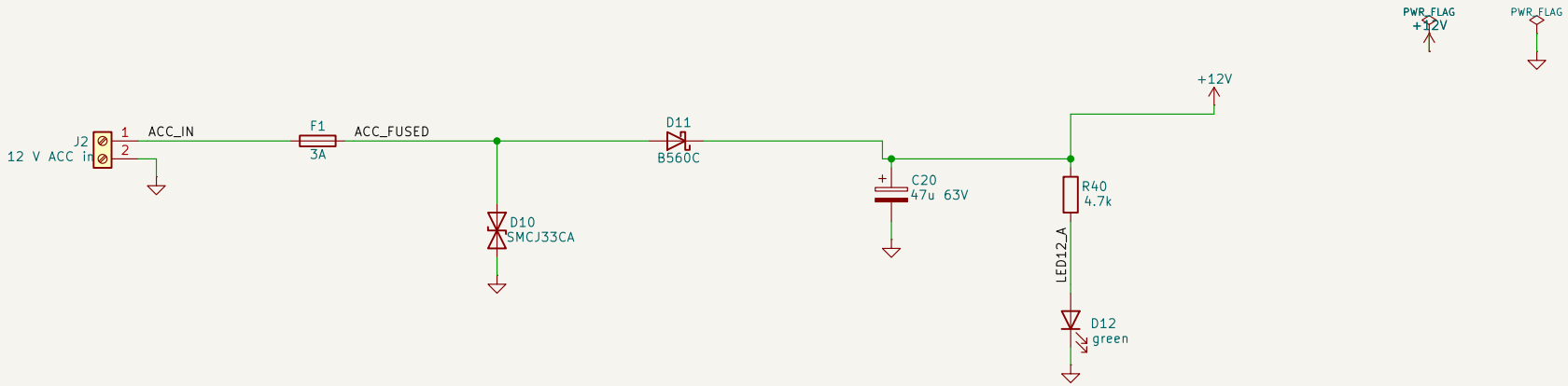
File: status.kicad_sch

12 V input -> 5 V converter -> 3.3 V and the pod's 5 V.
The pod drives the actuator, relays and CAN;
the status sheet reports back to it.

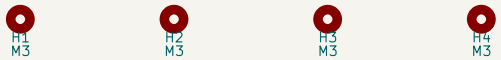
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP - Kiogora Ian Mwenda

Sheet: /		
File: power_board.kicad_sch		
Title: Under-dash power board (power, actuator, relays, CAN)		
Size: A3	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 1/9

12 V INPUT (the logic feed): switched ACC from the car's fuse box, so the board is off with the ignition. F1 3 A fuse. D10 bidirectional TVS clamps load-dump and jump-start spikes to about 39 V. D11 Schottky (60 V, 5 A) blocks a reversed lead. D12: +12V present.



Mounting holes: M3, no electrical connection



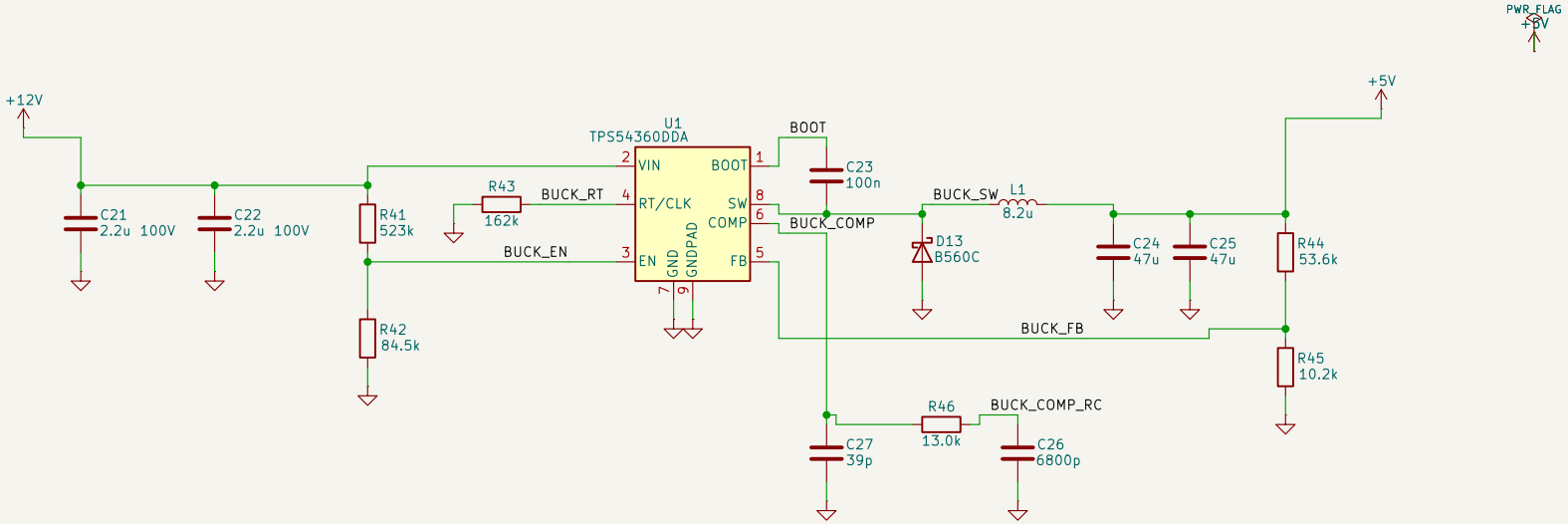
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /12 V input and protection/
File: input.kicad_sch

Title: 12 V input and protection

Size: A4	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 2/9

5 V: TI's TPS54360 5 V / 3.5 A reference design (datasheet SLVSBB4G, Figure 34), unchanged.
Rated to 60 V input, so whatever the TVS lets through a load dump cannot destroy it.
R41/R42 start it at 8 V and stop it at 6.25 V. R43 sets 600 kHz. R44/R45 set 5.0 V.
C23 is the bootstrap for the high-side switch; D13 carries the current while it is off.
R46, C26, C27 compensate the control loop. Lay out C21/C22, U1, D13 and L1 as one tight loop.



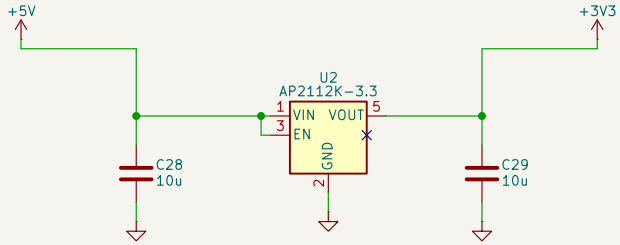
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP — Kiogora Ian Mwenda

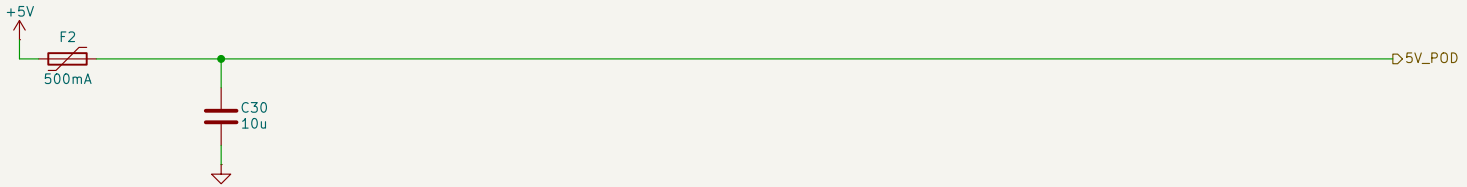
Sheet: /5 V converter (TPS54360, TI reference design)/
File: buck.kicad_sch

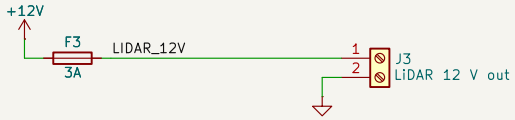
Title: 5 V converter (TPS54360, TI reference design)

Size: A4	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 3/9

3.3 V (AP2112K) for the CAN transceiver only; the pod makes its own 3.3 V.
5V_POD: the pod's supply up the cable, through F2, a 500 mA resettable fuse,
so a short in the cable cannot take down this board's 5 V.
LiDAR: 12 V through its own 3 A fuse F3 to the M12 socket.







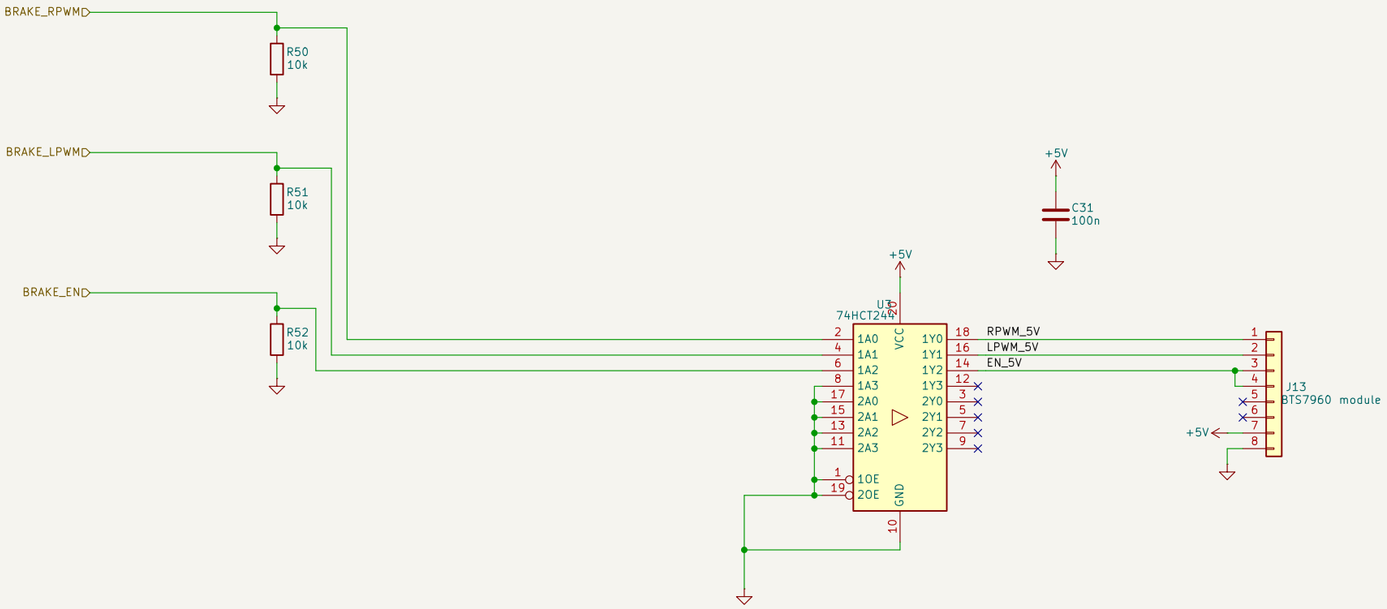
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /3.3 V, the pod's 5 V, the LiDAR feed/
File: rails.kicad_sch

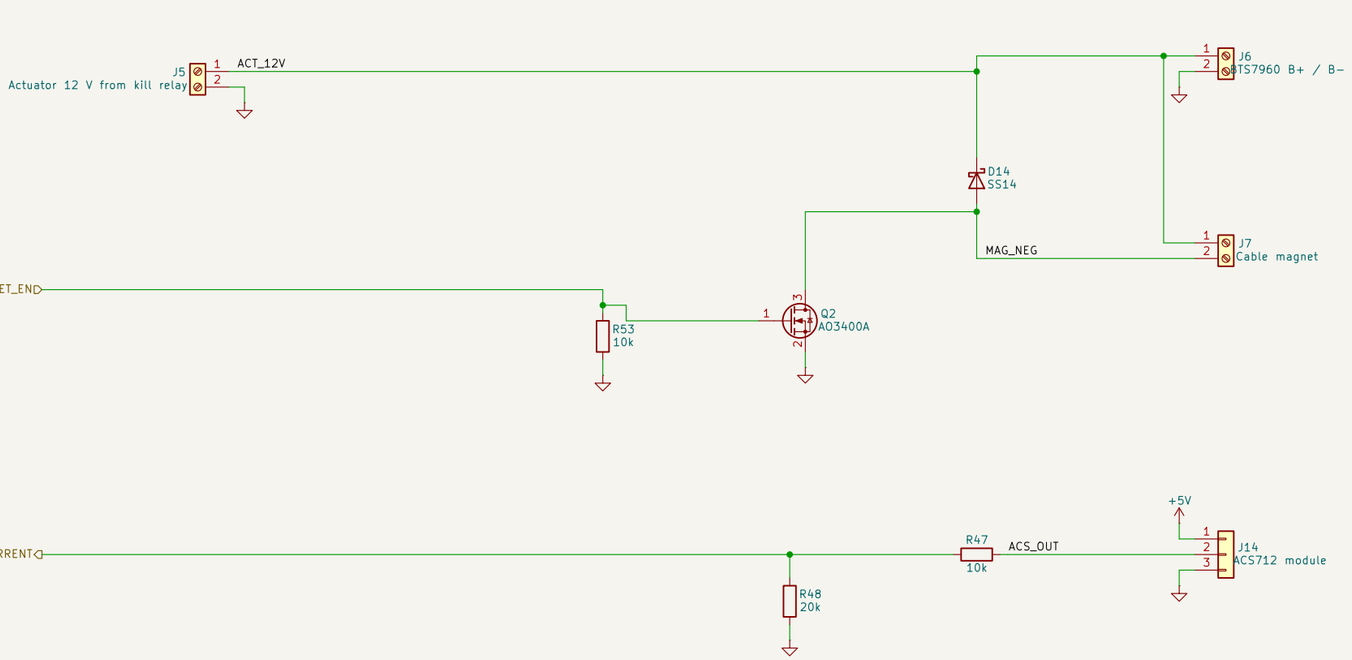
Title: 3.3 V, the pod's 5 V, the LiDAR feed

Size: A4	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 4/9

BRAKE ACTUATOR. Every line from the pod has a 10 k pull-down here (R50–R53): with the pod in reset, unpowered or unplugged, the drive, enable and magnet lines read LOW: released, cable dropped free. U3 (74HCT244, TTL thresholds) turns the pod's 3.3 V logic into the 5 V the BTS7960 module expects.



Actuator power comes ONLY through the 40 A kill relay's contacts (J5), whose coil is fed through the E-stop. Q2 holds the cable magnet; D14 absorbs its turn-off spike.



ACS712-20A module: 2.5 V at 0 A, 100 mV/A. R47/R48 scale it by 2/3 so nothing above 3.3 V travels up the cable to the pod's ADC.

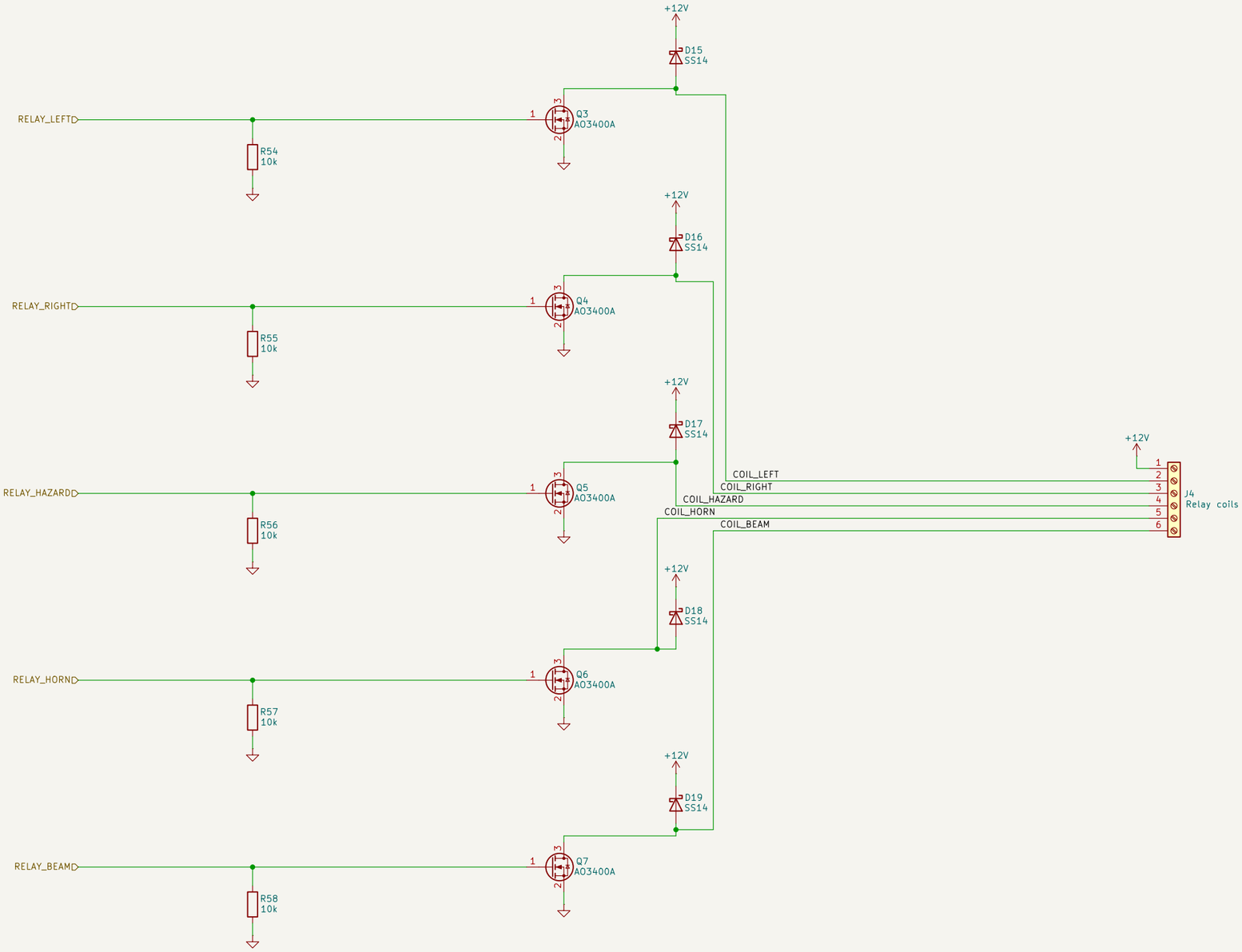
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /Brake actuator: buffer, BTS7960, current sense, cable magnet/
File: actuator.kicad_sch

Title: Brake actuator: buffer, BTS7960, current sense, cable magnet

Size: A3	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 5/9

SIGNAL RELAYS: five 12 V automotive relays in sockets, off the board; their contacts tap the car's switch wiring (Part 15.2). Each coil is switched low-side by a MOSFET whose gate has a 10 k pull-down (R54–R58): no drive from the pod, no relay. D15–D19 absorb each coil's turn-off spike.



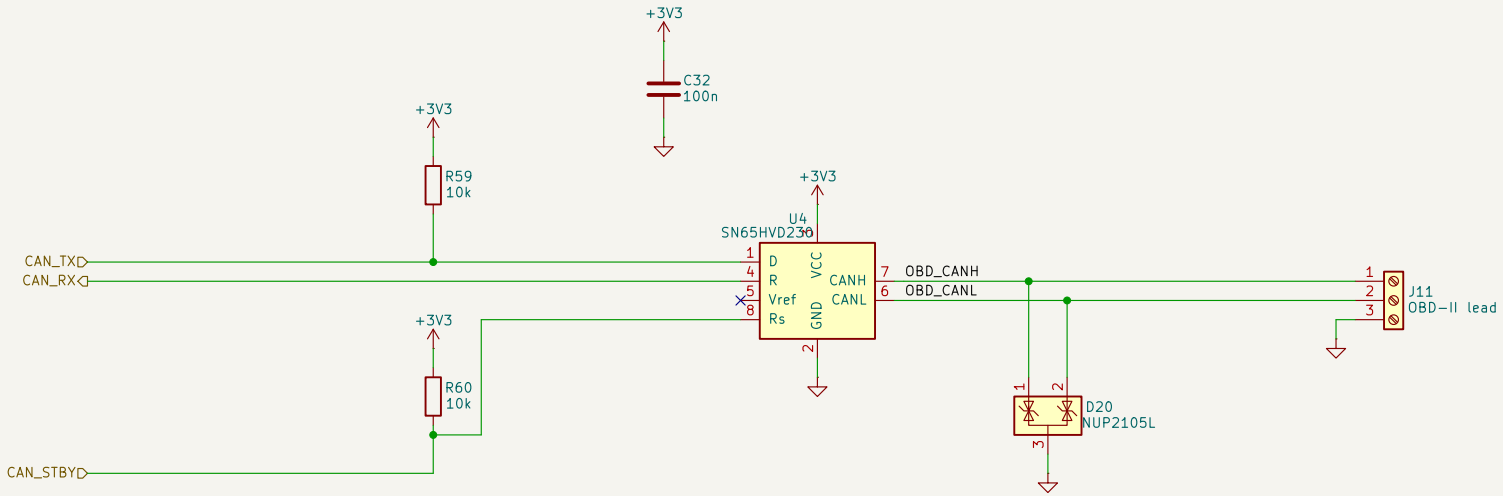
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /Signal relays: indicators, hazards, horn, high beam/
File: relays.kicad_sch

Title: Signal relays: indicators, hazards, horn, high beam

Size: A3	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 6/9

CAN (ISO 15765-4) to the car's OBD-II port, on this board so the unterminated branch off the car's bus is only the ~0.5 m OBD lead. R59 holds TX recessive and R60 holds Rs high (standby, cannot transmit) whenever the pod is not driving them. No termination: the car has its two.



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Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP — Kiogora Ian Mwenda

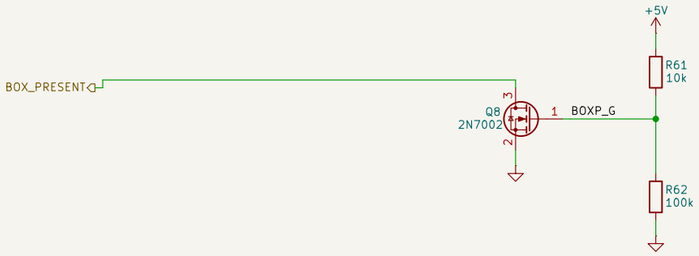
Sheet: /CAN transceiver for OBD-II/
File: can.kicad_sch

Title: CAN transceiver for OBD-II

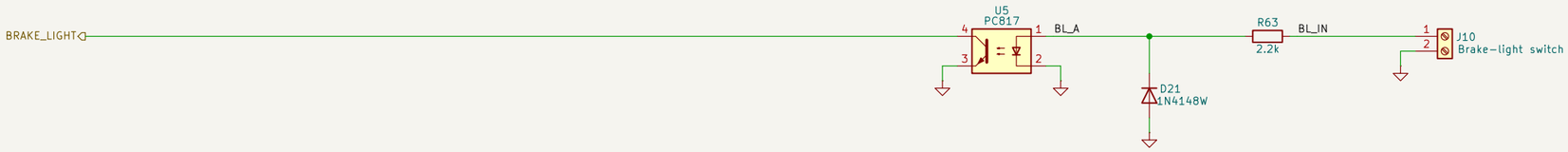
Size: A4	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 7/9

Inputs that report to the pod. Car side on the right, pod side (the ports) on the left.

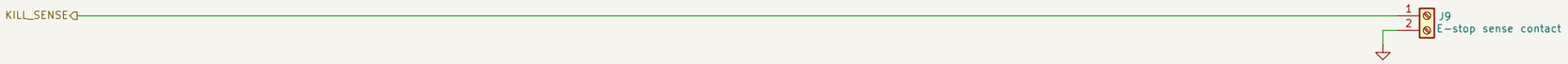
BOX_PRESENT: Q8 pulls it low only while this board's 5 V is up,
so the pod sees 'present' only when the box is connected AND powered.



Brake light: the opto-coupler U5 isolates the car's 12 V brake-light
circuit; D21 protects its LED from reverse voltage.



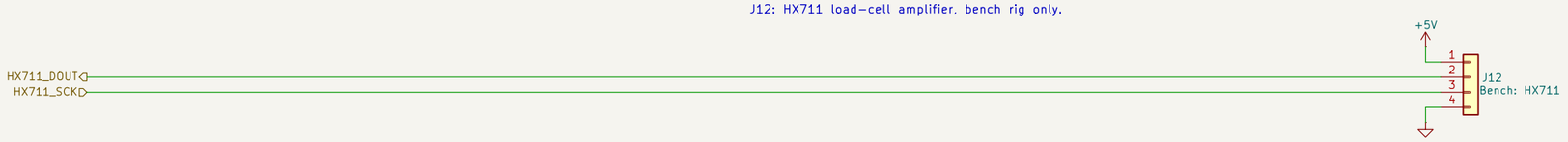
E-stop second contact (NC): released pulls KILL_SENSE low.



J8: +12V -> E-stop NC contact -> 40 A kill relay coil -> GND (wired off the board).



J12: HX711 load-cell amplifier, bench rig only.



Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
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Sheet: /Presence, brake light, E-stop sense, kill relay, bench/
File: status.kicad_sch

Title: Presence, brake light, E-stop sense, kill relay, bench

Size: A3	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 8/9

INTER-BOX CABLE (BUILD_GUIDE 4.8.3). Shield bonded to GND at this end only.
Every line driven by the pod has its fail-safe pull-down on the sheet that uses it.



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Under-dash power board (power, actuator, relays, CAN)
AR Windscreen FYP — Kiogora Ian Mwenda

Sheet: /Inter-box cable: DB-25 to the pod/
File: cable.kicad_sch

Title: Inter-box cable: DB-25 to the pod

Size: A4	Date: 2026-10-07	**Rev: 1**
KiCad E.D.A. 10.0.1		Id: 9/9